

SHETH L.U.J & SIR M.V COLLEGE
PRACTICALNO3

Aim: Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity

CODE :

```
import hmac
import hashlib
```

```
# Input
```

```
key = input("Enter Secret Key: ").encode()
message = input("Enter Message: ").encode()
```

```
# Generate MAC
```

```
mac = hmac.new(key, message, hashlib.sha256).hexdigest()
```

```
print("\nGenerated MAC:")
print(mac)
```

```
# Verification
```

```
print("\n--- MAC Verification ---")
received_message = input("Enter Message for Verification: ").encode()
received_mac = input("Enter MAC: ")
```

```
calculated_mac = hmac.new(key, received_message, hashlib.sha256).hexdigest()
```

```
if hmac.compare_digest(calculated_mac, received_mac):
```

```
    print("\nMAC Verified!")
    print("Message is Authentic and Unchanged.")
```

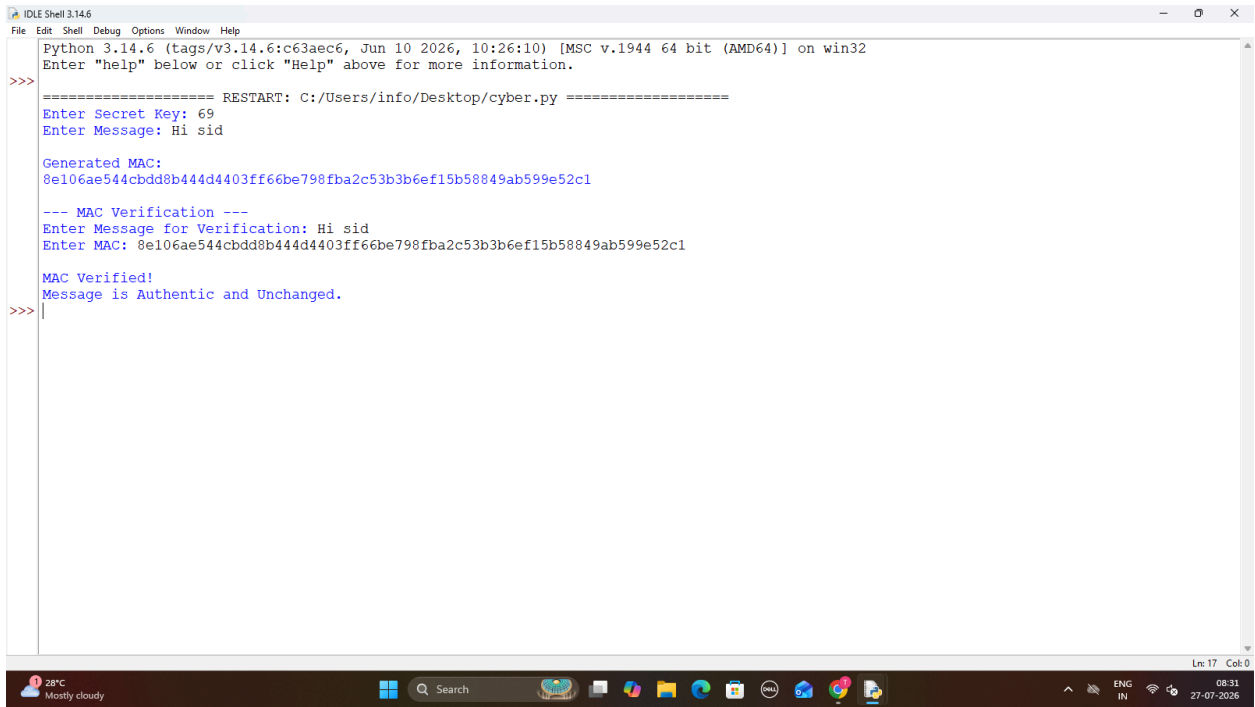
```
else:
```

```
    print("\nMAC Verification Failed!")
    print("Message has been Modified or MAC is Invalid.")
```

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OUTPUT:



```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/info/Desktop/cyber.py =====
Enter Secret Key: 69
Enter Message: Hi sid

Generated MAC:
8e106ae544cbdd8b444d4403ff66be798fba2c53b3b6ef15b58849ab599e52c1

--- MAC Verification ---
Enter Message for Verification: Hi sid
Enter MAC: 8e106ae544cbdd8b444d4403ff66be798fba2c53b3b6ef15b58849ab599e52c1

MAC Verified!
Message is Authentic and Unchanged.
>>>
```

GUI CODE:

```
import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

# Function to generate MAC
def generate_mac(key, message):
    return hmac.new(key.encode(), message.encode(), hashlib.sha256).hexdigest()

# Generate MAC Button
def generate():
    key = key_entry.get().strip()
    message = message_entry.get().strip()

    if not key or not message:
        messagebox.showerror("Error", "Secret Key and Message cannot be empty.")
        return

    mac = generate_mac(key, message)
```

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```
mac_entry.config(state="normal")
mac_entry.delete(0, tk.END)
mac_entry.insert(0, mac)
mac_entry.config(state="readonly")

# Verify MAC Button
def verify():
    key = key_entry.get().strip()
    received_message = verify_message_entry.get().strip()
    received_mac = verify_mac_entry.get().strip()

    if not key or not received_message or not received_mac:
        messagebox.showerror("Error", "Please fill all fields.")
        return

    calculated_mac = generate_mac(key, received_message)

    if hmac.compare_digest(calculated_mac, received_mac):
        messagebox.showinfo("Success", "MAC Verified!\nMessage is Authentic and Unchanged.")
    else:
        messagebox.showerror("Failed", "MAC Verification Failed!\nMessage has been Modified or the MAC is Invalid.")

# Main Window
root = tk.Tk()
root.title("Message Authentication Code (MAC)")
root.geometry("500x400")
root.resizable(False, False)

# Secret Key
tk.Label(root, text="Secret Key").pack(pady=5)
key_entry = tk.Entry(root, width=50, show="*")
key_entry.pack()

# Original Message
tk.Label(root, text="Message").pack(pady=5)
message_entry = tk.Entry(root, width=50)
message_entry.pack()

# Generate Button
tk.Button(root, text="Generate MAC", command=generate, bg="lightblue").pack(pady=10)
```

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Generated MAC

```
tk.Label(root, text="Generated MAC").pack()  
mac_entry = tk.Entry(root, width=60, state="readonly")  
mac_entry.pack()
```

Verification Section

```
tk.Label(root, text="Message for Verification").pack(pady=10)  
verify_message_entry = tk.Entry(root, width=50)  
verify_message_entry.pack()
```

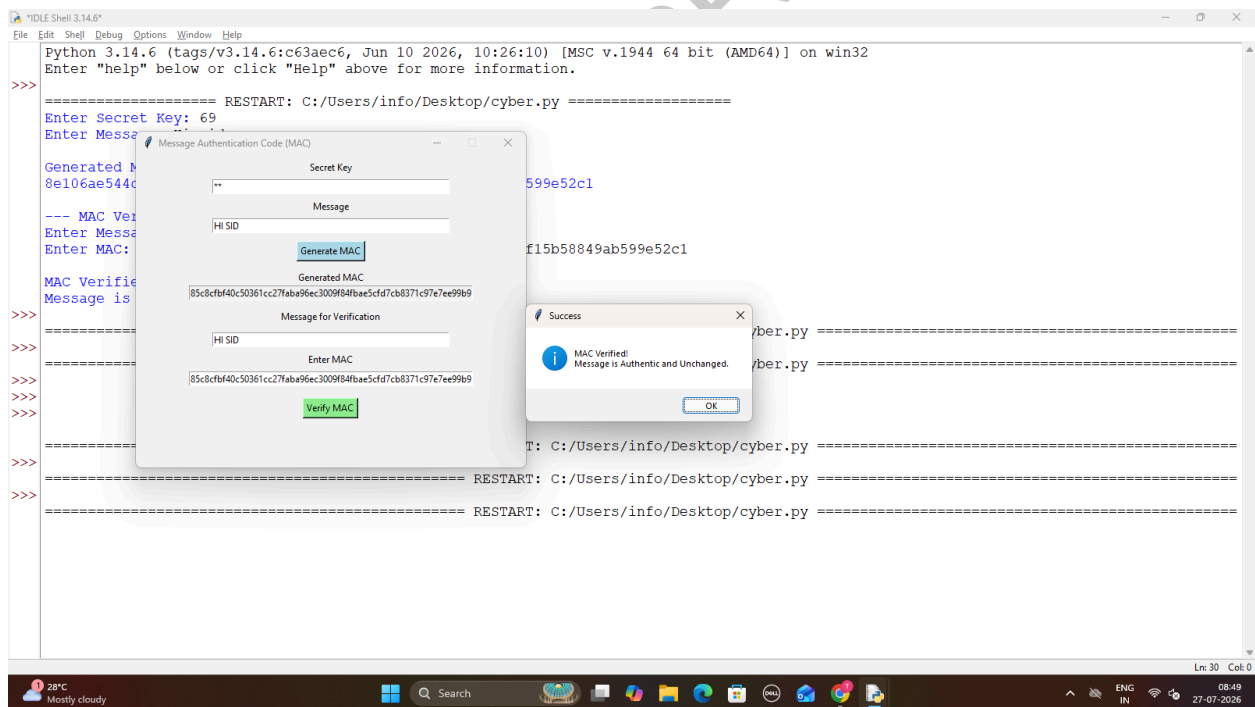
```
tk.Label(root, text="Enter MAC").pack(pady=5)  
verify_mac_entry = tk.Entry(root, width=60)  
verify_mac_entry.pack()
```

Verify Button

```
tk.Button(root, text="Verify MAC", command=verify, bg="lightgreen").pack(pady=15)
```

```
root.mainloop()
```

OUTPUT:



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